Precambrian Research, 3 (1976) 303—315

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Research Papers

PALEOMAGNETISM OF THE CHIBOUGAMAU GREENSTONE BELT, QUEBEC, AND THE EFFECTS OF GRENVILLIAN POST-OROGENIC UPLIFT*

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(Received March 3, 1975; accepted September 24, 1975)

ABSTRACT

Ueno, H. and Irving, E., 1976. Paleomagnetism of the Chibougamau greenstone belt, Quebec, and the effects of Grenvillian post-orogenic uplift. Precambrian Res., 3: 303—315.

A paleomagnetic survey (34 sites, 200 core samples) of an Archean greenstone belt just north of the Grenville Front is described. Sixty per cent of the samples have blocking temperatures less than 400° C and remanent coercive forces predominantly below 100 Oe, and they provide no information about the Precambrian geomagnetic field. The other samples contain many magnetizations of which two (CS and CH) can be clearly recognized. The CS magnetization occurs in basic sills, and has high blocking temperature (550 to 650°C) and remanent coercive force (often exceeding 1000 Oe) and a direction (186°,—38°) that is not significantly different from that in Matachewan diabase dikes (2690 m.y.). CS is thought to have been caused by uplift following the Kenoran orogeny and its age is probably about 2600 m.y. No older magnetization has been found. The CH magnetization occurs in several different bodies and has a direction (119°, +51°) similar to that observed in rocks from other places close to the Grenville Front. CH has more variable blocking temperatures (200 to 650°C) and remanent coercive forces (200 to > 1000 Oe) and is considered to have been acquired by heating during uplift of the highly metamorphose Grenville Province to the south. Uplift occurred about 1000 m.y. ago, and the CH magnetization is considered to be of this age. Evidence is presented to support the suggestion that there was an extensive magnetization episode both within and adjacent to the Grenville Province at that time.

INTRODUCTION

In this paper a paleomagnetic survey of an Archean greenstone belt is described. One of the disappointing features is the common occurrence of rocks with low blocking temperatures and low remanent coercive forces and which do not provide a record of the ancient geomagnetic field. This is not true of all the rocks, however, and some have retained remanent magnetizations of Precambrian age. The study area lies mainly in the Superior Structural Province

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immediately north of the Grenville Front (Fig.1), and one of the purposes of this work has been to study the effect on the remanent magnetization of the nearby uplift of highly metamorphosed Grenville rocks. The problem is illustrated in Figs.2 and 3. In Fig.2 the general distribution of paleomagnetic poles derived from Precambrian rocks at North America in the age range 1400—600 m.y. is shown. They fall into two groups. The poles from the Grenville Structural Province fall in the north Atlantic, and those from the rest of Laurentia (Interior Laurentia) near what is nowadays Africa. The two groups overlap in the central Atlantic and, as will be described below, many of the poles from this overlap zone have been obtained from rocks near to the Grenville Front. It is possible therefore that the magnetizations corresponding to the poles from the overlap zone are related in some way to the formation of the Grenville Front. In Fig.3 the distribution of K-Ar dates across the Grenville Front at Chibougamau is shown. In the Grenville Province the mean is just under 1000 m.y. In the Archean Superior Province as a whole the mean dates are 2500 m.y., but up to 20 km NW of the Front the dates are depressed, as if heat, derived from the uprising Grenville Province, drove out part of the

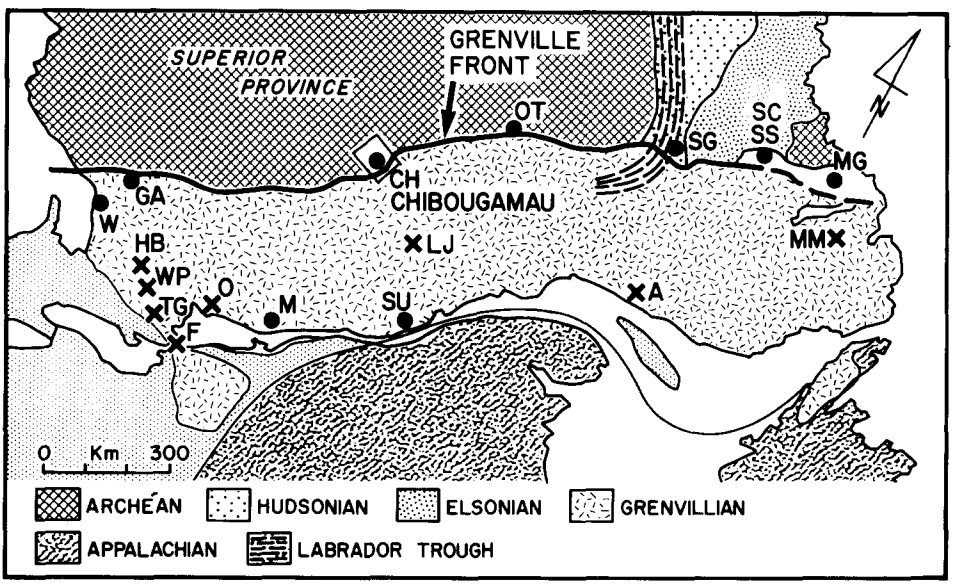


Fig.1. Paleomagnetic sampling localities in and near the Grenville Province. Results from localities marked by solid dots are discussed in this paper. They are as follows: *W*, Whitestone anorthosite and diorite (Ueno et al., 1975); *GA*, Grenville Front anorthosite (Palmer and Carmichael, 1973); *M*, Morin anorthosite (Irving et al., 1974); *SU*, St. Urbain anorthosite (Hargraves and Roy, 1974); *OT*, Otish gabbros (Fahrig and Chown, 1973); *SG*, Shabogamo gabbro (Fahrig et al., 1974); *SC*, Seal and Crouteau Groups igneous rocks (Roy and Fahrig, 1973); *SS*, Seal red-beds (Roy and Fahrig, 1973); and *MG*, Michael gabbro (Fahrig and Larochelle, 1972). Other localities are marked by crosses and are documented in Irving et al. (1974a,b). Results from these localities are not discussed here because either their magnetization is due to hematite or the mineral responsible for the magnetization has not yet been identified.

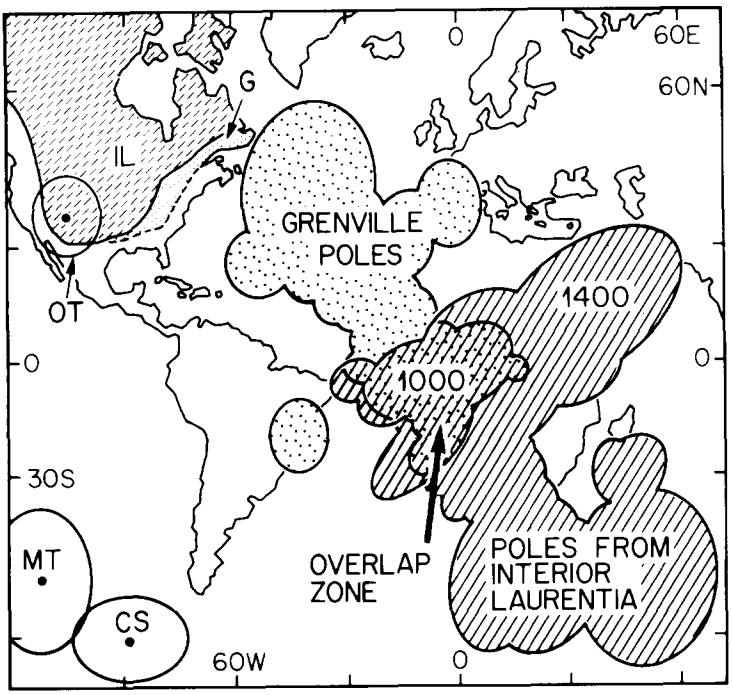


Fig.2. The distribution of poles from Laurentia in the age range −1400 to −600 m.y. The areas of the error circles only are shown. The individual poles are plotted and documented in Irving et al. (1974b). The poles from the Grenville Structural Province (G) and the rest of Laurentia (Interior Laurentia IL) have different ornament. Poles from Otish gabbro *OT* (Fahrig and Chown, 1973), Matachewan dykes *MT* (Strangway, 1964; Fahrig et al., 1965), and Chibougamau sills CS magnetization (Table II) are also marked.

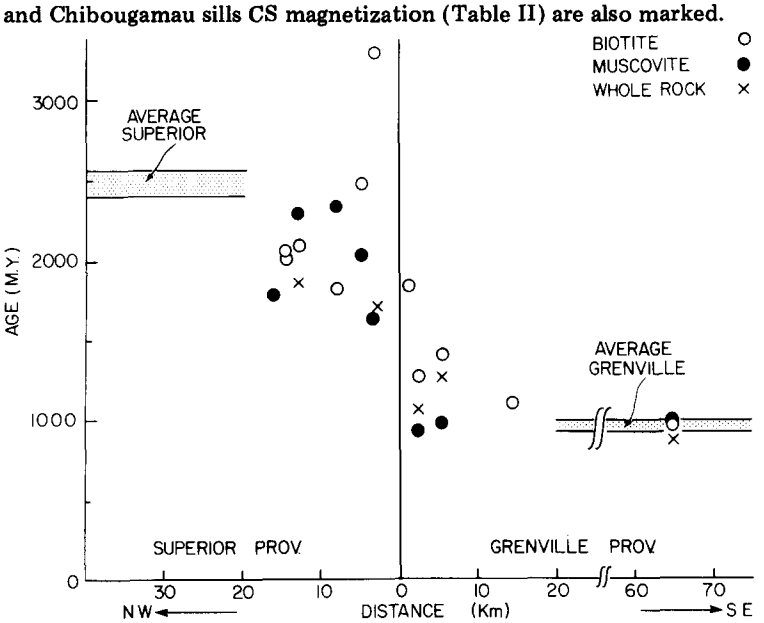


Fig.3. Changes in K-Ar dates across the Grenville Front near Chibougamau. Compiled from Lowdon (1961), Leech et al. (1963), Stockwell and Williams (1964), Wanless et al. (1965, 1966, 1968, 1970), Lowdon et al. (1968).

radiogenic argon and lowered the apparent age. The question therefore arises, could this heat have succeeded in resetting the directions of remanent magnetization into the direction of the field at the time of uplift and cooling, and thus have produced the overlap zone of Fig.2?

GEOLOGY OF CHIBOUGAMAU AREA

The Grenville Front is the boundary between the highly metamorphosed rocks of the Grenville Province to the south, and the less altered rocks of the Archean Superior Province to the north (Fig.1). The rocks of the Chibougamau greenstone belt were folded and metamorphosed to green-schist facies during the Kenoran orogeny (about 2600 m.y. ago) which occurred at the end of the Archean (Duquette, 1972). The oldest rocks are Keewatin-type volcanic rocks, called the Roy Group (Fig.4). They have not been sampled. The Roy Group has been invaded by seven intrusive units which are now given in order of decreasing age.

Metagabbro sills. These bodies, which are too small to show in Fig.4, have been sampled at site 64 (medium-grained hornblende-chlorite metagabbro) and site 83 (fine-grained pyroxene metagabbro).

Doré Lake Complex. This is a layered intrusion varying from ultrabasic to silicic in composition (Allard, 1972). It has a total thickness of over 4000 m. An Rb/Sr isochron date of 2629± 173 m.y. (decay constant $1.39 \cdot 10^{-11}$ y^{-1}) has been reported from the complex (Jones et al., 1974). The units present are as follows: (a) upper border zone, not sampled; (b) sodagranophyre zone, not sampled; (c) ferrodiorite zone, sites 83 and 84; (d) layered zone, site 67 (pyroxenite), site 68 (gabbroic anorthosite), sites 69, 71, 73, 85 and 86 (pyroxenite and gabbro with oxide layers), and site 70 (gabbroic anorthosite); (e) anorthosite zone, site 72 (anorthositic gabbro) and sites 56, 57 and 58 (recrystallized anorthosite).

Roberge sill. This is about 1000 m thick, and consists of serpentized dunite and peridotite (site 54).

Ventures sill. This is about 1000 m thick, and consists of a lower medium-grained pyroxenite (site 79), and an upper coarse-grained gabbro (sites 53, 78, 80, and 81).

Bourbeau sill. This is about 700 m thick. Site 51 is in pyroxenite, and site 52 is leucogabbro.

Tonalite suite. This consists of tonalite and diorite, and is considered to have been emplaced immediately prior to the Kenoran folding (Allard, 1972). An Rb/Sr isochron date of 2563± 79 m.y. has been reported by Jones et al. (1974).

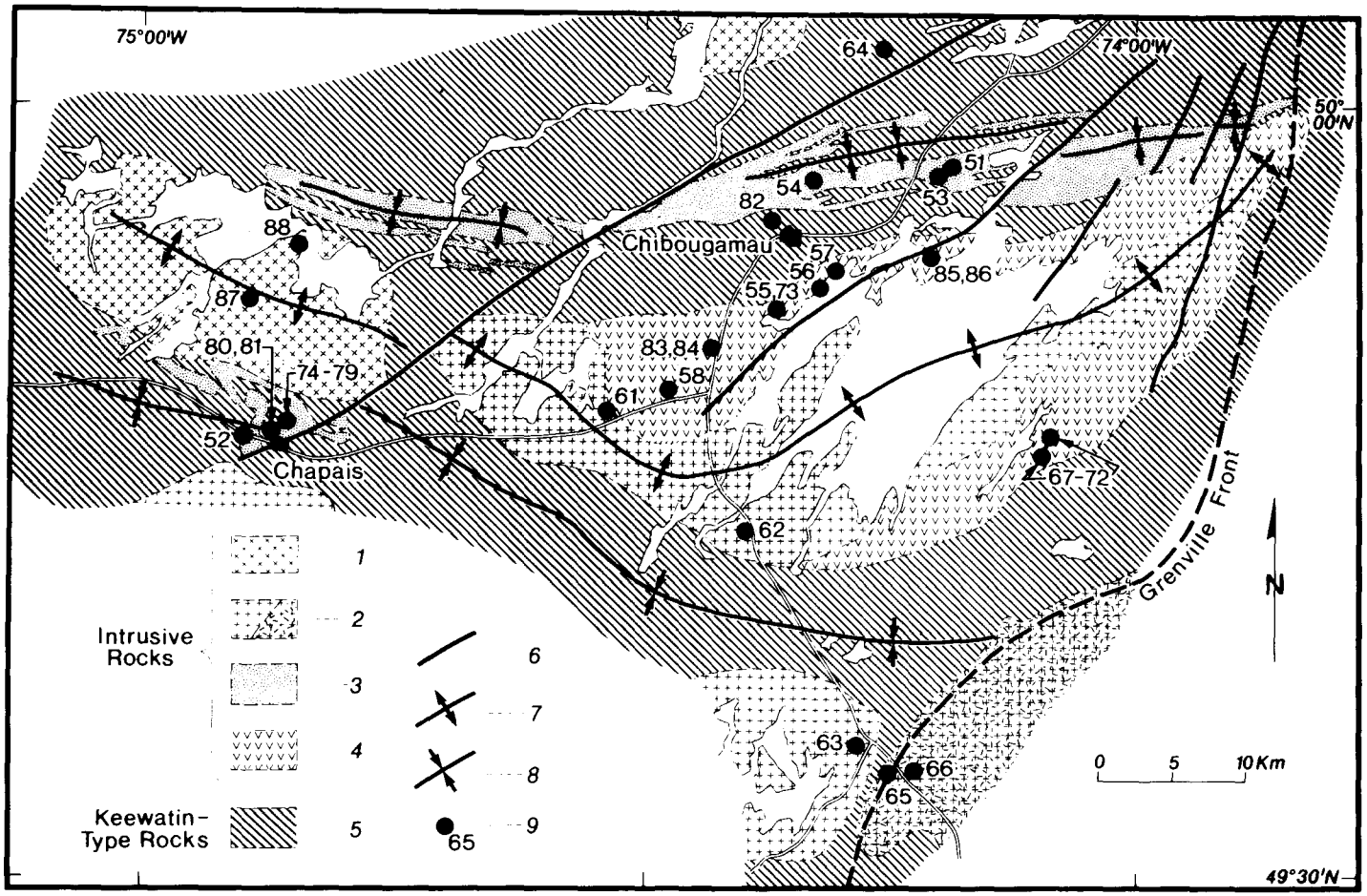


Fig.4. Geological sketch map of Chibougamau area compiled from these sources Graham (1956), Derry and Folinsbee (1957), Horscroft (1958), Allard (1960, 1972), Smith (1960), Duquette, (1964, 1972). The legend is as follows: *1*, granodiorite suite; *2*, tonalite suite; *3*, Roberge, Ventures, and Bourbeau sills; *4*, Doré Lake Complex; *5*, Roy Group; *6*, fault; *7*, anticline; *8*, syncline; *9*, sampling sites.

Samples were collected from sites 61, 62, and 63. Gneissic samples from metamorphosed equivalents of the tonalite suite within the Grenville Province were collected at sites 65 and 66.

Granodiorite suite. This is considered to be post-Kenoran, but has not been dated. It has been sampled at sites 87 and 88.

The metamorphism of the Chibougamau rocks probably occurred soon after formation. The area is a general synclinorium, comprising two synclines separated by an anticline (Fig.4). Folding is steep, the dips of sills are commonly 60° or more, and they are occasionally overturned.

PALEOMAGNETIC RESULTS

Samples were collected from 34 sites. Five cores were obtained at each site, and two specimens were cut from each core. Remanent magnetization was measured by a Schonstedt SM1 spinner magnetometer. Thermal and alternating field (a.f.) demagnetization was carried out in quasi-zero field ($< 2\gamma$) using the apparatus described by Roy et al. (1969, 1971, 1973).

Thermal demagnetization has been carried out on 25 specimens and the decay curves are of four types (*A*—*D* in Fig.5). The *A*-type curves are square-shouldered, and the blocking temperatures range from 550 to 650° C. Iron oxides of the *A*-type specimens observed in polished sections are hematite, magnetite, and ilmenite. The gneissic rock at site 66 within the Grenville Province has blocking temperatures about 650° C, and contains rod-shaped hematite which is a recrystallization product.

The *B*-type curves have more widely distributed blocking temperatures, ranging from 400 to 600° C. The iron oxide minerals are magnetite (1—100 μ) with associated hematite often rimming the magnetite. The *C*- and *D*-type curves have much lower blocking temperatures, generally less than 350° C. Large magnetite crystals with large ilmenite lamellae are common. Sulphides are abundant. The recrystallized anorthosite (site 56) contains mostly sulphides. At site 87, where the blocking temperatures are below 250°C, the rock contains large magnetite grains (100—200 μ) replaced in part by hematite.

Alternating field demagnetization curves were obtained from one specimen from each site. The magnetization in most sites is very unstable, there are no end-points, and the directions in specimens from the same site are random. A typical result is shown in Fig.6(a). This random magnetization (referred to as **CR** magnetization) occurs at 20 sites (one site in the metagabbros, ten in the Doré Lake Complex, four in the sills, and five in the tonalite and granodiorite suites). Altogether 60% of the samples collected were magnetically unstable. The magnetization at the remaining fourteen sites shows a varying degree of stability, and the results of a.f. demagnetization studies are summarized in Table I. Three magnetizations (**CS, CV** and **CH**) have been recognized.

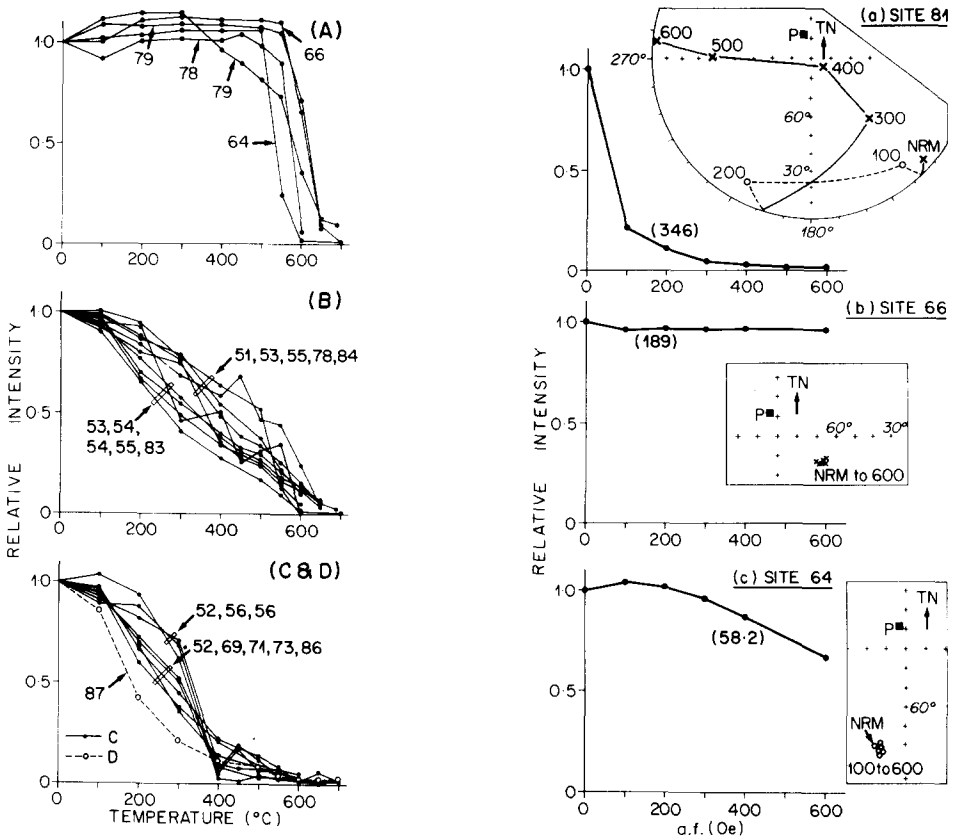


Fig.5. Thermal demagnetization curves. The intensity is normalized to unity at room temperatures. The four types (*A*—*D*) are described in the text. Each curve is labelled by site number. When site numbers are repeated a curve from more than one sample from that site has been obtained.

Fig.6. Alternating field demagnetization. Decay curves and corresponding direction changes are shown in the insets which are parts of stereograms. Circles denote negative inclination, crosses positive inclination. The initial (n.r.m.) intensity in mG is given in brackets. (a) Typical random CR magnetization. (b) Tonalite with very stable CH magnetization. (c) Gabbro with stable CS magnetization.

The CS magnetization has a southerly direction and occurs at four sites. It is very stable (Fig.6(c)) being characterized by *A*-type thermal decay curves. The CS magnetization is found only in the sills. The results are summarized in Table II. The standard deviation of the directions in situ is 10°, but if the directions are corrected assuming the sills to be horizontal, this increases to 57°. Hence the magnetization post-dates the folding. The pole position agrees with that for the Matachewan dikes (Fig.2) which are dated at 2690±93 m.y. (Gates and Hurley, 1973). It is probable therefore that the magnetization was acquired during cooling and uplift following Kenoran orogeny at about the same time as the extensive swarm of Matachewan dikes was introduced. CS

TABLE I

Site results

Site	$N(n)$	a.f. (Oe)	$D°, I°$	k	Thermal type	Magnetization
Granodiorite suite						
87	5	200	348,+84	50	*D*	CX
Tonalite suite (metamorphosed in Grenvillian orogeny)						
66	5	300	129,+55	43	*A*	CH
Bourbeau sill						
51	4(1)	200, 300	137,+45	14	*B*	CH
Ventures sill						
53	5(2)	200	190,—46	11	*B*	CS
78	5(1)	200	067,—28	21	*A,B*	CV
79	5(2)	200	054,—20	24	*A*	CV
78′	3(2)	200, 300	179,—32	12	*A*	CS
79′	3(2)	2000	183,—30	25	*A*	CS
Roberge sill						
54	5(2)	200, 300	116,+72	41	*B*	CH
Doré Lake Complex						
73	5	n.r.m.	357,—35	26	*C*	CX
86	5	300	219,+22	95	*C*	CX
69	4(1)	300	028,+39	93	*C*	CX
70	5	200	090,+54	24	*B*	CH
71	4(1)	400	323,+36	10	*C*	CX
72	4(1)	300	117,+23	144	*B*	CH
Metagabbro sill						
64	2(3)	400	194,—44	928	*A*	CS

N is the number of cores used in the analysis, and n is the number discarded because specimen directions from them differed by $> 50°$; a.f. is the cleaning field used; D, I are the declination and inclination of the mean direction; and k is the precision (Fisher, 1953). The thermal and magnetization types are described in the text.

TABLE II

Summary

Magnetization (sites)	B	$D°, I°$	k	Pole	dm, dp
CS (53, 64, 78′, 79′)	4	186,—38	70	61S,087W	13,08
CH (51, 54, 66, 70, 72)	5	119,+ 51	15	07S,155E	28,19
CV (78, 79)	2	060,—24	64	08S,134W	34,18

B is the number of sites. In the two right-hand columns the pole position and semi-angles of error ellipse are given. Other symbols as in Table I. Both CS and CV magnetizations occur at sites 78 and 79 and are separated by a.f. cleaning as explained in text.

appears to be the oldest magnetization present, and its age is probably about 2600 m.y.

The CH magnetization has a southeasterly direction, and was found at five sites. It occurs in tonalite gneiss just inside the Grenville Province (site 66), where it is extremely stable, with coercivities in excess of 1000 Oe, and *A*-type thermal decay curves (Figs.5, and 6(b)). The other occurrences of **CH** are in the Doré Lake Complex and the sills (sites 51, 54, 70 and 72). They are characterized by *B*-type thermal decay curves, and remanent directions coercivities of about 300 Oe. The standard deviation of the directions in situ is 21° (k=21), but if the directions are corrected assuming the sills and igneous layering to be horizontal this increases to 42° (k=4), indicating that **CH** is post-Kenoran. Its origin is considered later.

At two sites in the Ventures sill there is northeasterly magnetization referred to as CV. Thermal decay curves are always of the *B*-type. The CS and **CV** magnetizations usually occur in separate specimens from the Ventures sill, but occasionally they are found in the same specimen. In these the directions are initially parallel to CV and remain there in a.f. fields of up to 200 Oe, but migrate to the CS direction after treatment in 2000 Oe. CS always has a higher remanent coercive force than CV. CV is not widespread, and has a pole position (Table II) close to that expected for rocks of about 1500 m.y. old. It could therefore be an overprint of about that age.

In addition to these three magnetizations there are magnetizations which, although reasonably well-grouped within a site, show no agreement between sites (Table I). These are referred to collectively as **CX** magnetizations. Their decay curves are always of *C*- or *D*-type, that is, they have low blocking temperatures, so that their remanence could have been reset by mild heating over extended intervals. Such mild heating may have occurred repeatedly during the history of the Chibougamau area. Hence their magnetization has probably been reset at different times, and this presumably accounts for their variability. No attempt is made to interpret the **CX** magnetizations individually. All can be matched approximately with known post-Archean field directions. For example site 87 in the granodiorite suite has a direction close to that observed from rocks at about 2200 m.y. (Pullaiah and Irving, 1975). The directions at site 69 in the Doré Lake Complex could be a reversed CS magnetization, but it has not been grouped with them because it has a *C*-type magnetization thermal decay curve.

DISCUSSION

We now consider the CH magnetization and kindred magnetizations found elsewhere. At Chibougamau CH occurs in Archean rocks north of the Grenville Front, and also in high-grade gneisses immediately south of the Front (site 66, Fig.4). The same magnetization direction occurs elsewhere both inside, and marginal to, the Grenville Province. Inside the province it is characteristic of rocks whose remanence is due to magnetite (Whitestone anortho-

site *XY* magnetization, Morin anorthosite *M*, and the St. Urbain anorthosite *SU*, see Figs.1 and 7). Marginal to the province it has been found both at the Front itself (Grenville Front anorthosites *GA*) and immediately north of the Front (Michael gabbros *MG*, igneous rocks of the Seal and Croteau Groups *SC2*, and the Shabogamo gabbros *SG2*). The corresponding poles, which are referred to as zone I in Fig.7, fall in the central Atlantic, within the overlap zone of Fig.2.

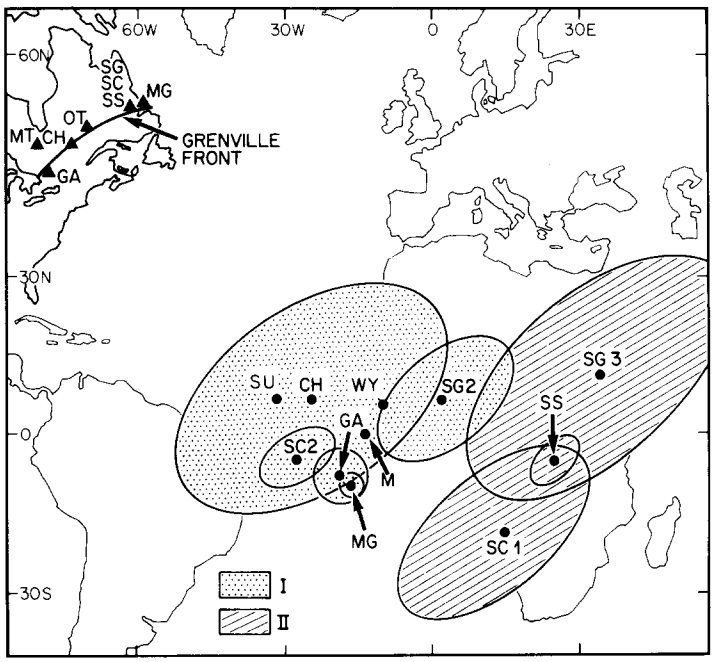


Fig.7. Poles from near Grenville Front. The symbols are listed in the legend to Fig.1. The error circles are at P=0.05. Poles from localities OT and MT are plotted in Fig.1. Poles from magnetite remanences observed deep within the Grenville Province are added for comparison: *WY*, *M*, and *SU*.

Two of these results are of special interest. The igneous rocks of the Seal and Croteau Groups have two magnetizations with different directions and poles (Roy and Fahrig, 1973). The dominant magnetization (*SC2*) is of moderate remanent coercivity (about 200 Oe). The less common magnetization (*SC1*) has a very high coercivity (over 2,000 Oe) and is found in reddish hematitic basalts that were presumably oxidized soon after extrusion. Both magnetizations sometimes occur in the same specimens and can be separated by a.f. demagnetization (Roy and Fahrig, 1973; Fig.7e). The pole *SC1* does not differ significantly from the well-defined pole for the red beds of the Seal Lake Group (*SS*) which have a remanence with high-blocking temperatures (680° C) and which is carried by hematite (Fig.7). The distinction is analogous to that which we observe at Chibougamau, where rocks just north of the Front have two

magnetizations, one with moderate (CH), and the other (CS) with high coercivities and blocking temperatures. Likewise the results from the Shabogamo gabbro (Fahrig et al., 1974) may be arranged into two: one group (with pole *SG2*) from rocks within 30 km of the Front, and another (*SG3*) from rocks at 80 km from the Front (Fig.7).

When these distinctions are made the poles from rocks in the vicinity of the Grenville Front fall into two zones (Fig.7). Zone I comprises poles from rocks of moderate coercivity within 20 or 30 km north of the Front (*SC2*, CH, *MG*), from hematite remanence in rocks from immediately south of the Front (*GA* and site 66), and from magnetite remanence deep within the Grenville Province (*SU*, *M*, *MY*). Zone I is in the overlap zone of Fig.2. Poles either from hematite remanences near to the front or from rocks at greater distances fall in what is referred to as zone II (*SC1*, *SS*, *SG3*). The poles of zone II are obtained from rocks whose ages lie between 1300 and 1400 m.y., and fall close to the poles from rocks of this age from elsewhere in the Laurentian Shield (see discussion and diagrams in Irving et al., 1974b). Hence their corresponding magnetizations are probably primary. We suggest that the poles of zone I were acquired at a time of a widespread magnetization episode that occurred as a result of Grenville uplift. Deep within the Grenville Province itself, this magnetization was acquired during uplift and cooling. Immediately north of the Front the magnetite remanence (CH, *SC2*, *MG*, *SG2*) was reset more or less completely at approximately the same time by heat diffusing from the rising Grenville Province to the south, forming a "magnetite magnetic aureole" north of the Front. Hematite magnetizations (*SS*, CS, *SC1*) within this aureole, because of their higher blocking temperatures, were unaffected. That there must have been a thermal aureole north of the Front very probably caused by Grenville uplift is shown by the K-Ar ages which up to 18 km north of the Front are reset downwards by about 500 m.y. (Fig.3). Partial or complete resetting of magnetizations within this zone is therefore to be expected, and appears to have occurred.

If this argument is correct then the mean of the zone I poles provides an estimate of the pole position relative to Laurentia at the time of Grenville uplift. The mean is 0°N,018°W (N=8, k=41, A_{95}=9°). The age we tentatively assign to this pole is 1000 m.y., which is the age of Grenville uplift at the Front itself determined from K-Ar dates from biotite. One difficulty in this explanation is that the K-Ar ages trend from about 1000 m.y. at the front itself to about 850 m.y. along the St. Lawrence (Harper, 1967). If this reflects differential uplift, earlier near the Front, then a trend of poles rather than a single group might be expected; the agreement of poles in zone I is perhaps better than might be expected. However, zone I covers a substantial area, and as more accurate data are obtained, systematic polar trends may be found within it.

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